

Welcome to COGS 108!

Data Science in Practice

Shannon E. Ellis, Ph.D

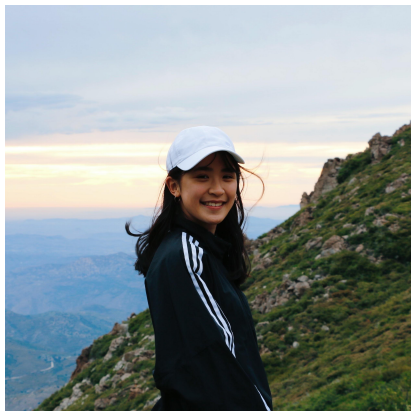


Teaching Professor
Cognitive Science Department
[Halicioğlu Data Science Institute](#)

Lectures: <https://github.com/COGS108/Lectures-Ellis/tree/fa26>



COGS 108 Instructional Staff: TAs



Zoe



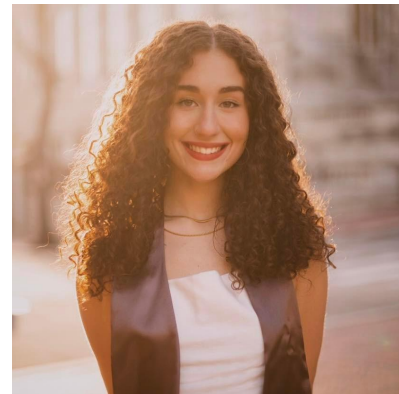
Niranjan



Marlyn



Chaitanya



Liz

+ Reader: Jaisurya

The (dreaded) waitlist

1. I know this matters to you and is a source of stress (and I hate that).
2. I do not have access to the system to let someone off of the waitlist
- 3. I don't anticipate anyone getting off the waitlist this quarter**
 - a. Project Feedback is crucial in this course. Each TA grades ~22 projects.
 - b. If students drop (not common), staff will enroll students from the waitlist
 - c. Our staff (cogsadvising@ucsd.edu) take care of this
4. The waitlist settles after week 2.

Students on the waitlist have access to Canvas/all course materials. If you want/hope to be enrolled, please keep up with the material these first two weeks. **You will be expected to have completed the material if you are eventually enrolled.**

...so what exactly is data science?



what is biology

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The study of life, living things, & organisms



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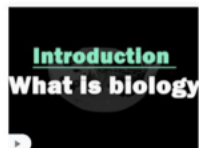
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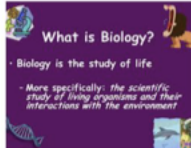
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living



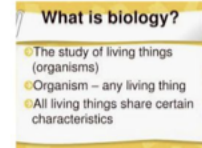
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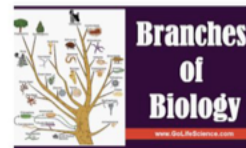
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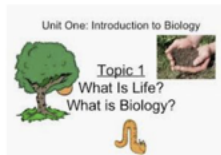
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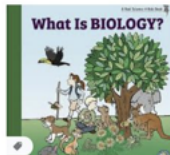
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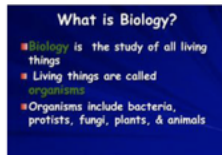
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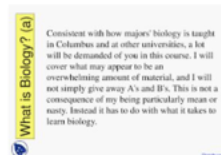
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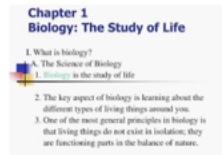
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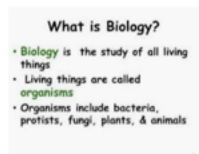
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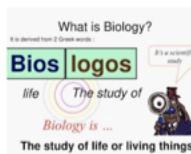
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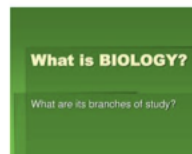
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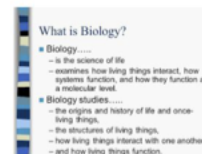
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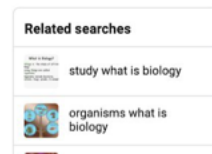
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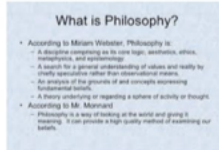
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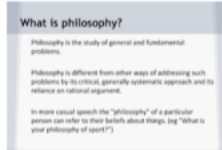
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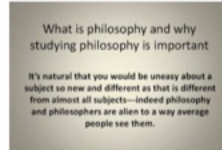
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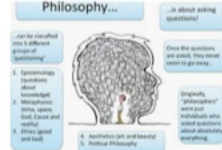
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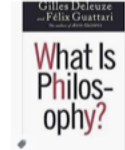
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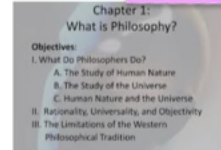
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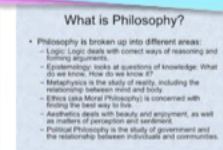
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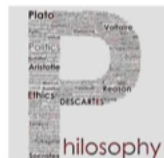
by Gilles Deleuze, Felix...
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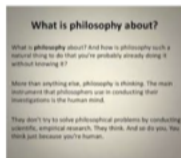
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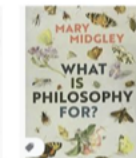
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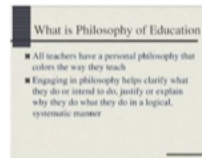
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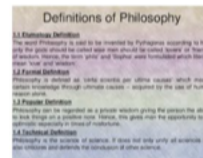
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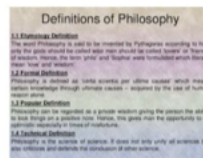
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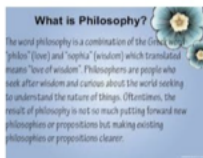
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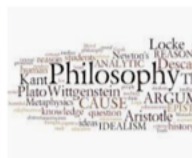
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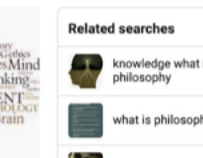
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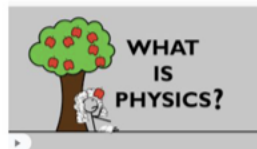
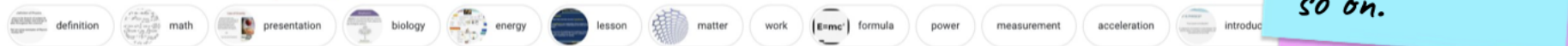
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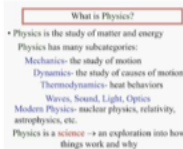
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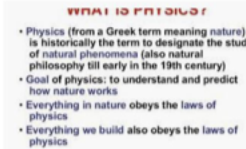
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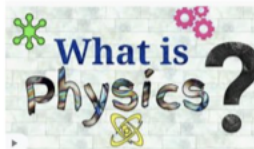
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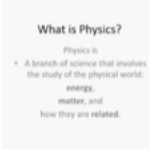
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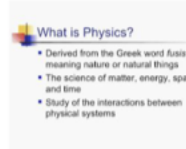
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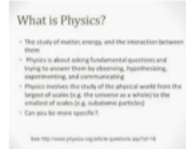
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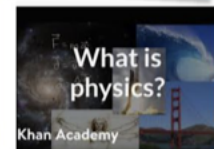
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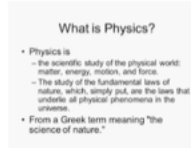
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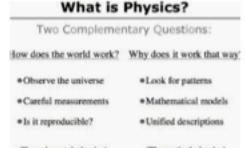
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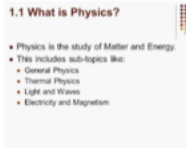
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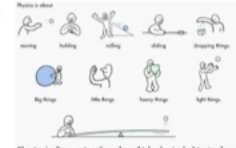
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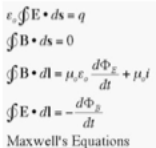
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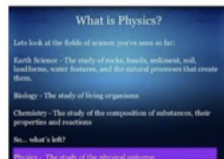
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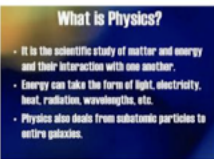
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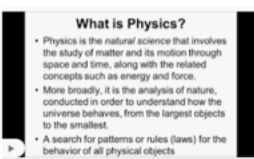
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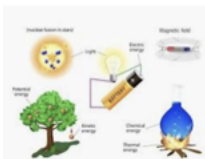
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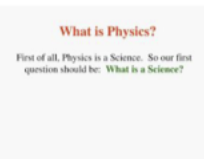
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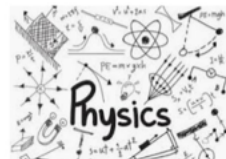
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Study of
matter, energy,
atoms,
mechanics, and
so on.



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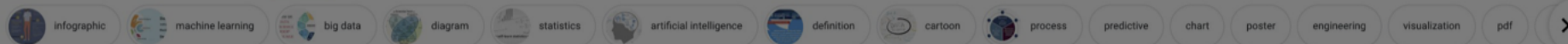
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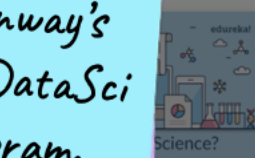
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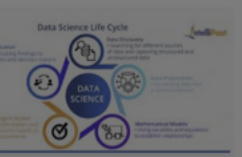
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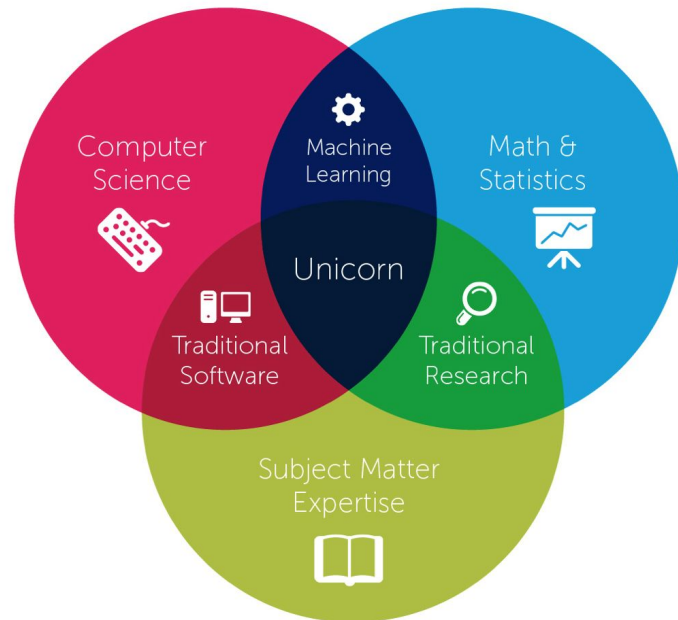
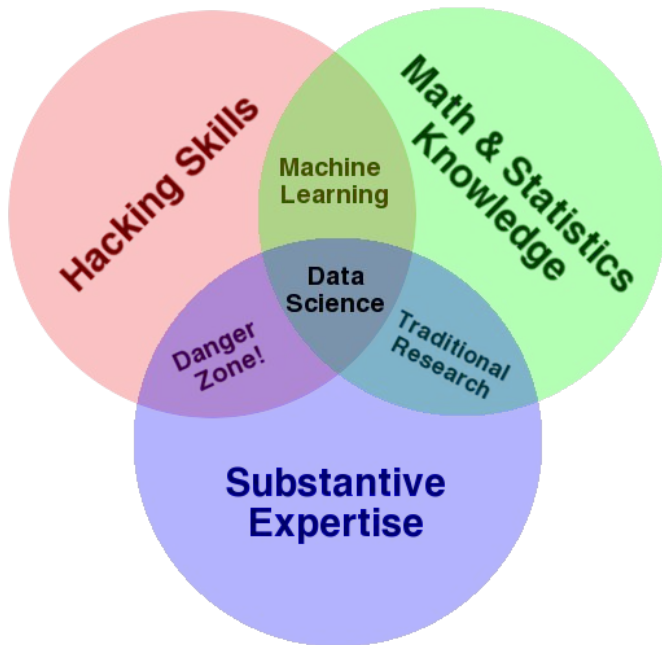
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Drew Conway's
original DataSci
venn diagram.
(The OG)

What is data science?



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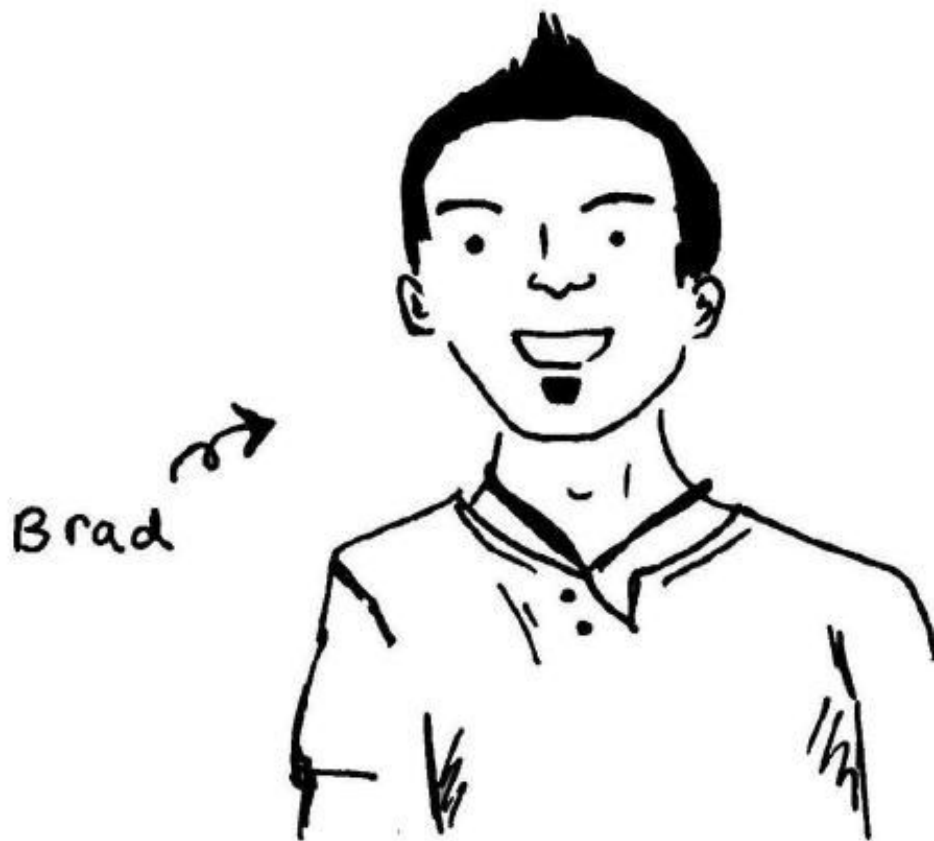
David Donoho

This coupling of scientific discovery and practice involves the collection, management, processing, analysis, visualization, and interpretation of vast amounts of heterogeneous data associated with a diverse array of scientific, translational, and interdisciplinary actions.



Bradley Voytek

The study of how the quantification of observable phenomena can lead to human understanding of the processes giving rise to those phenomena—or even the ability to predict future outcomes absent human understanding—and why certain phenomena require more or less data to lead to human understanding and/or prediction accuracy



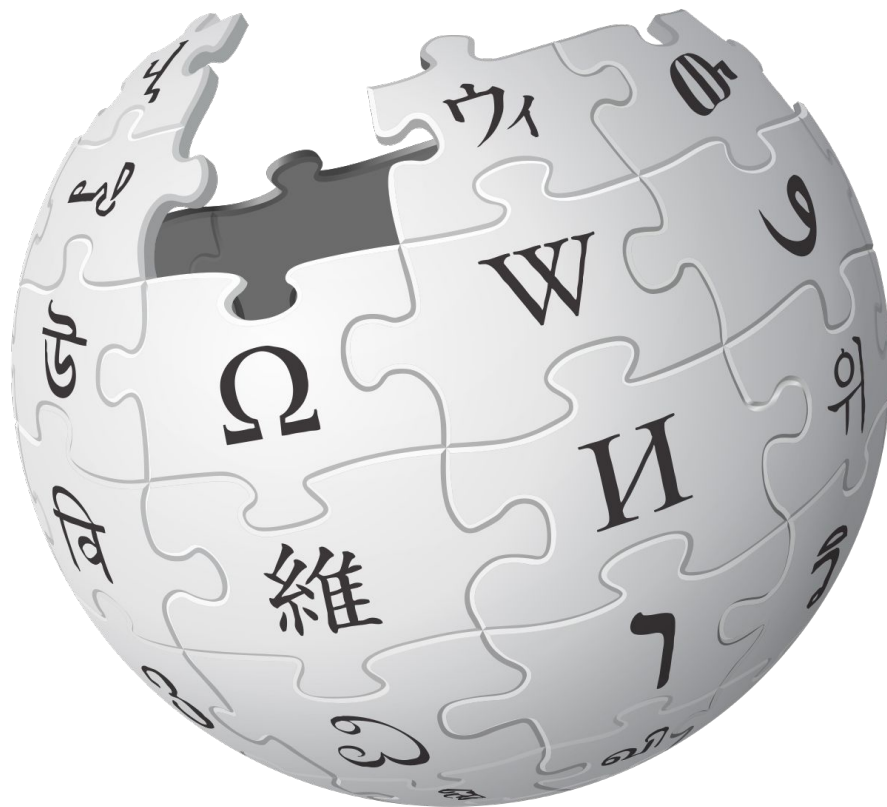
Rafael Irizarry

An umbrella term used by organizations to describe the processes used to extract value from data.



Wikipedia

A concept to unify statistics, data analysis, machine learning and their related methods" in order to "understand and analyze actual phenomena" with data.^[3] It employs techniques and theories drawn from many fields within the context of [mathematics](#), [statistics](#), [information science](#), and [computer science](#).



Cogs 108 Definition

The scientific and ethical process of extracting value from data.



Fastest Growing Occupations

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Fastest growing occupations: 20 occupations with the highest projected percent change of employment between 2025–35.

Click on an occupation name to see the full occupational profile.

OCCUPATION	GROWTH RATE, 2025–35	2025 MEDIAN PAY
Nurse practitioners	41%	\$132,300 per year
Solar photovoltaic installers	37%	\$53,140 per year
Data scientists	35%	\$120,230 per year
Wind turbine service technicians	30%	\$64,120 per year
Medical and health services managers	24%	\$123,860 per year

**Data scientists ask
interesting questions &
answer them with data**

The goal in COGS 108 is to *do* data science.

Course Objectives

- Formulate a plan for and complete a data science project from start (question) to finish (communication)
- Explain and carry out descriptive, exploratory, inferential, and predictive analyses in Python
- Communicate results concisely and effectively in reports and presentations
- Identify and explain how to approach an unfamiliar data science task

How we'll approach
learning about *and*
doing data science in
COGS 108

Scheduling

Lecture: MWF (1P @ PCYNH 106 ; 2P @ PETER 110)

Discussion Sections: M, W

Prof Office Hours:

- Tu 1-3 (zoom, by appt)
- W 3-4 (CSB 243, drop-in)

Section	Day	Time	Location	Room	Staff
001-001	M	11:00a-11:50a	PCYNH	120	Liz
001-002	M	12:00p-12:50p	PCYNH	120	Liz
001-005	M	3:00p-3:50p	CSB	004	Zoe
001-006	M	4:00p-4:50p	CSB	004	Zoe
002-001	W	9:00a-9:50a	DIB	122	Marlyn
002-002	W	10:00a-10:50a	PCYNH	120	Marlyn
002-003	W	11:00a-11:50a	PCYNH	120	Niranjan
002-004	W	12:00p-12:50p	PCYNH	120	Niranjan
002-005	W	1:00p-1:50p	PCYNH	120	Chaitanya
002-006	W	3:00p-3:50p	PCYNH	120	Chaitanya

What are course staff up to in COGS 108?

TAs | leading discussion section, project feedback + grading, answering questions, providing me with feedback (student-facing)

Reader | managing autograding, grade entry, participation tracking

When helping, our goal is to do just that. Help. Often, you'll be under time pressure to complete an assignment (which we understand), but we *really* care about your learning. So, we all will prioritize guiding you to an answer to ensure learning (rather than just giving an answer away).

COGS 108: General Plan

Week	Topic(s)
1	Version Control & Data
2	Data Intuition, Wrangling, Ethics
3	DS Questions & Data Visualization
4	Exploratory Data Analysis & Inference
5	Dimensionality Reduction & ML
6	Machine Learning
7	Geospatial Analysis
8	Text Analysis & Communication
9	How to Be Wrong
10	Guest Lectures + Oral Exams

Programming Prerequisite

- MAE 8 - MATLAB
- CSE 8A or 11 - Python/Java
- COGS 18 - Python
- DSC 10 - Python
- (CSS 1 - Python)

Bottom line: we will assume programming knowledge.
Python will be used for all projects and assignments.

No programming experience (or you forget it all)?

- *Preferred option*
 - Take a programming course first
 - COGS 18 : Introduction to Python
- *Can't wait?*
 - Use online sites like [codecademy.com](https://www.codecademy.com) or [LearnPython.org](https://www.learnpython.org)
 - [Python Data Science Handbook](#)

Course links

GitHub	https://github.com/COGS108	lecture/section slides & final projects
datahub	https://datahub.ucsd.edu	assignment completion & submission
Q&A (Piazza)	https://piazza.com/class/mublhuuzdqr5v8 <i>(or access directly from Canvas)</i>	questions, discussion, and regrade requests
Canvas	https://canvas.ucsd.edu/courses/77704	Deadlines, grades
Anonymous Feedback	Submit via Google Form	if I ever offend you, use an example you hate, or to provide general feedback that you don't want your name attached to

	% of Total Grade
Pre-course Survey (#finalid)	1
(9/10) Weekly Quizzes (lecture content)	9
Participation (or final exam)	25
(9) Assignments (technical)	27
Final Group Project	38
(1) Project Review*	2
(1) Project Proposal*	5
(2) Project Checkpoints*	10
(1) Oral Exam	7
(1) Final Report*	10
(1) Final Video*	3
(1) Project Survey	1

* indicates group submission

Participation is tracked

Google Forms will be used for in-class lecture Q&A and activities and discussion participation

- Participate in 30/35 lectures/sections
 - **Lectures** (27) : Weeks 1-9 MWF + 2 “guest lectures” in week 10 - 2 holidays
 - **Discussion** (8): Weeks 2-6; 8-10
- To get credit for attendance, you must be present and participate in a given day’s activities (submit the Google Form(s))
 - watching the podcast does not count

Notes:

- you can attend *either* lecture on a given day to get credit. If you attend both, you’ll only get credit for attending one day’s lecture.
- For Google Forms, you’ll have to use your UCSD email for login
- We will work to update you on your participation weekly

Participation or final exam (25%)

If you participate: no final exam

Why? I'm confident that you can get what you need out of this course without a final exam, *if* you engage with the material throughout

I would LOVE if everyone chose participation! Think of this as the default.

If you don't participate: a (reasonably difficult) in-person, closed notes final exam

...if you don't participate, I need to measure that you have learned/met the objectives of the course

Why Google Forms in COGS 108?

- There are a whole lot of you!
- Checks understanding
- Provides me with feedback
- Aids in critical thinking & allows for application of concepts
- Give you all a break from listening to me (we humans need this!)

Weekly Lecture Quizzes:

- (9/10) weekly quizzes (first one due Sunday of week 1)
- Goal: to help you keep on top of the material covered in lecture
- Why?: experience + student feedback
- How:
 - Taken on Canvas
 - Single Attempt
 - Open notes, open internet, closed other people
 - ~10 Questions
 - Timed : 15 minutes
 - Posted by Friday @ 5PM (after each week of lecture)
 - Meant to test concepts from that week's lecture

Lecture quizzes will be due on Sundays by 11:59 PM.

No late submissions. Lowest quiz score will be dropped.

All deadlines Sun at 11:59 PM

Week	Quiz	Lecture Quiz	Assignment	Project
1	Version Control & Data	Q1	--	
2	Data Intuition, Wrangling, Ethics	Q2	A1	Group Sign-up*
3	DS Questions & Data Visualization	Q3	A2	Project Review*
4	Exploratory Data Analysis & Inference	Q4	A3	Project Proposal*
5	Dimensionality Reduction & ML	Q5	A4	
6	Machine Learning	Q6	A5	Checkpoint 1: Data*
7	Geospatial Analysis	Q7	A6	
8	Text Analysis & Communication	Q8	A7	Checkpoint 2: EDA*
9	How to Be Wrong	Q9	A8	
10	Oral Exams + Guest Lectures	Q10	A9	Final report*, video*, evaluation
11	Final Exam (M or W)			

*indicates group submission. All other assignments/surveys are completed & submitted individually.

(9) Assignments

Assignments are completed individually and graded programmatically.

- These are meant to get you practice programming around the topics covered in class.
- The first two are much simpler than the following two and should take less time.
- You will have to look some stuff up on your own. This is by design.
- Instructions must be followed to receive credit.

Assignments will be due on Sundays by 11:59 PM.

7 free late days; 75% credit if submitted w/n 5d after deadline after that.
Applied automatically.

All deadlines Sun at 11:59 PM

Week	Quiz	Lecture Quiz	Assignment	Project
1	Version Control & Data	Q1	--	
2	Data Intuition, Wrangling, Ethics	Q2	A1	Group Sign-up*
3	DS Questions & Data Visualization	Q3	A2	Project Review*
4	Exploratory Data Analysis & Inference	Q4	A3	Project Proposal*
5	Dimensionality Reduction & ML	Q5	A4	
6	Machine Learning	Q6	A5	Checkpoint 1: Data*
7	Geospatial Analysis	Q7	A6	
8	Text Analysis & Communication	Q8	A7	Checkpoint 2: EDA*
9	How to Be Wrong	Q9	A8	
10	Oral Exams + Guest Lectures	Q10	A9	Final report*, video*, evaluation
11	Final Exam (M or W)			

*indicates group submission. All other assignments/surveys are completed & submitted individually.

Assignment Submission @ Datahub: <https://datahub.ucsd.edu>

DATA SCIENCE / MACHINE LEARNING PLATFORM

UC San Diego

Information Technology Services - Educational Technology Services

Help Options ▾



Log In

Registered Users
"username@ucsd.edu"

UC San Diego Jupyterhub (Data Science) Platform

Before Fri: log onto datahub. (If you have Canvas access, you have datahub access)

Group Projects: the main focus of COGS 108

Groups of 4-5 Individuals

Carry out the entire data science process on a topic of your choosing

All deadlines Sun at 11:59 PM

Week	Quiz	Lecture Quiz	Assignment	Project
1	Version Control & Data	Q1	--	
2	Data Intuition, Wrangling, Ethics	Q2	A1	Group Sign-up*
3	DS Questions & Data Visualization	Q3	A2	Project Review*
4	Exploratory Data Analysis & Inference	Q4	A3	Project Proposal*
5	Dimensionality Reduction & ML	Q5	A4	
6	Machine Learning	Q6	A5	Checkpoint 1: Data*
7	Geospatial Analysis	Q7	A6	
8	Text Analysis & Communication	Q8	A7	Checkpoint 2: EDA*
9	How to Be Wrong	Q9	A8	
10	Oral Exams + Guest Lectures	Q10	A9	Final report*, video*, evaluation
11	Final Exam (M or W)			

*indicates group submission. All other assignments/surveys are completed & submitted individually.

Discussion Section

Goals:

- Discuss important DS readings
- Make project progress (work with groupmates; feedback from TA/grader)
- Get questions answered
- Interact with other humans

There will be Google Form(s) to complete in section each week. These will be used for your participation.

Discussion Sections will start week 2; no section week 7 (holiday)

Project: Group Formation

You will have time to form groups within section during week 2.

If you choose to form groups outside of your assigned section, EVERYONE in your group must have:

- At least two weekly sections you are all able to attend together
- A single lecture slot when you're all able to attend in week 10

Course Confusion

- If something in lecture, an assignment, instructions, etc. is unclear:
 - *ask in class*
 - *ask during section*
 - *post on Piazza*
 - *ask a classmate*
 - *come to office hours*

I look at Piazza first every day. Then email. I have ~600 students this quarter. Please use Piazza when possible. (I do answer email and Canvas messages, but please spare me.)

Academic Integrity Policy

Don't cheat.

Group work is required (projects) and encouraged throughout to help you learn.

COGS 108 is meant to be collaborative. We have a lot of students. Let's harness that and learn from one another.

But, copying + pasting someone's work (including from GenAI) without understanding is not learning.

Work type	Group work / ask other students	Open book / internet	Ask a clarification question to TA / professor	GenAI usage
Exam	✗	✗	✓	✗
Lecture Quizzes	✗	✓	✓	⚠
Assignments	✓	✓	✓	⚠
Project	✓	✓	✓	⚠

✓ permitted

✗ not allowed

⚠ indicates that use is permitted; however, GenAI should never be asked to complete a quiz, assignment, or project component for you. Instead you should be using it conversationally, with targeted specific tasks and/or questions.

**A message for
first-gen students,
transfer students,
and those who don't
have older
siblings/friends who
have attended
UCSD/university**

If you are struggling, come to office hours. Ask questions on Piazza. Reach out to me to ask for better approaches. Your classmates ARE doing this. And, you're not alone.

If you need a bit longer on something b/c you fell sick, a family thing came up, work called you in for an extra shift, etc., you have 7 free late days.

If something takes you beyond the 7 free late days, we can chat. Your classmates ARE doing this.

**What COGS 108 logistics
questions do you have?**

Why this course?

You are going to be analyzing lots of data because you're studying to be a:

cognitive scientist

data scientist

computer scientist

neuroscientist, biologist, or chemist

social scientist (linguist? psychologist?)

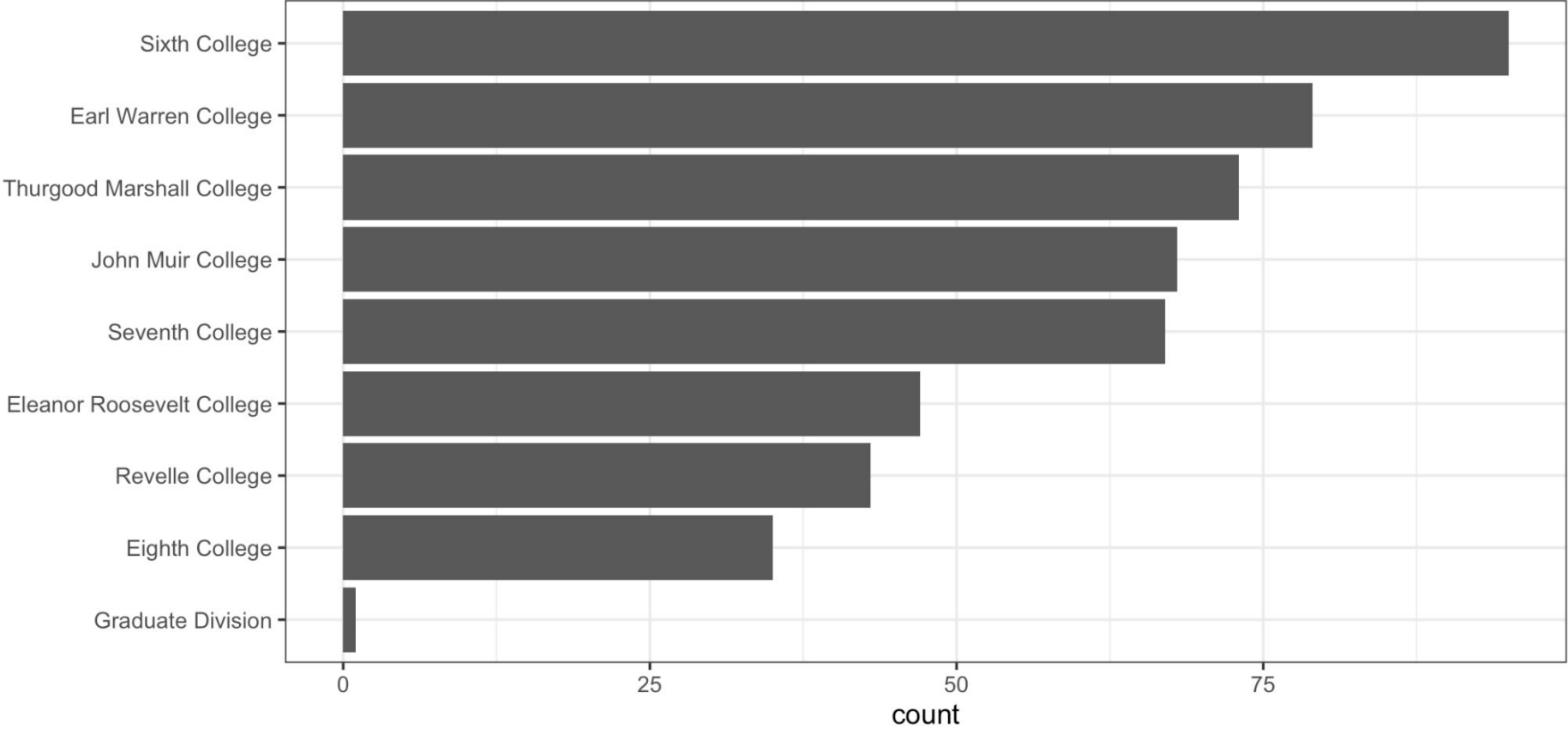
statistician or biostatistician

CEO/small business owner

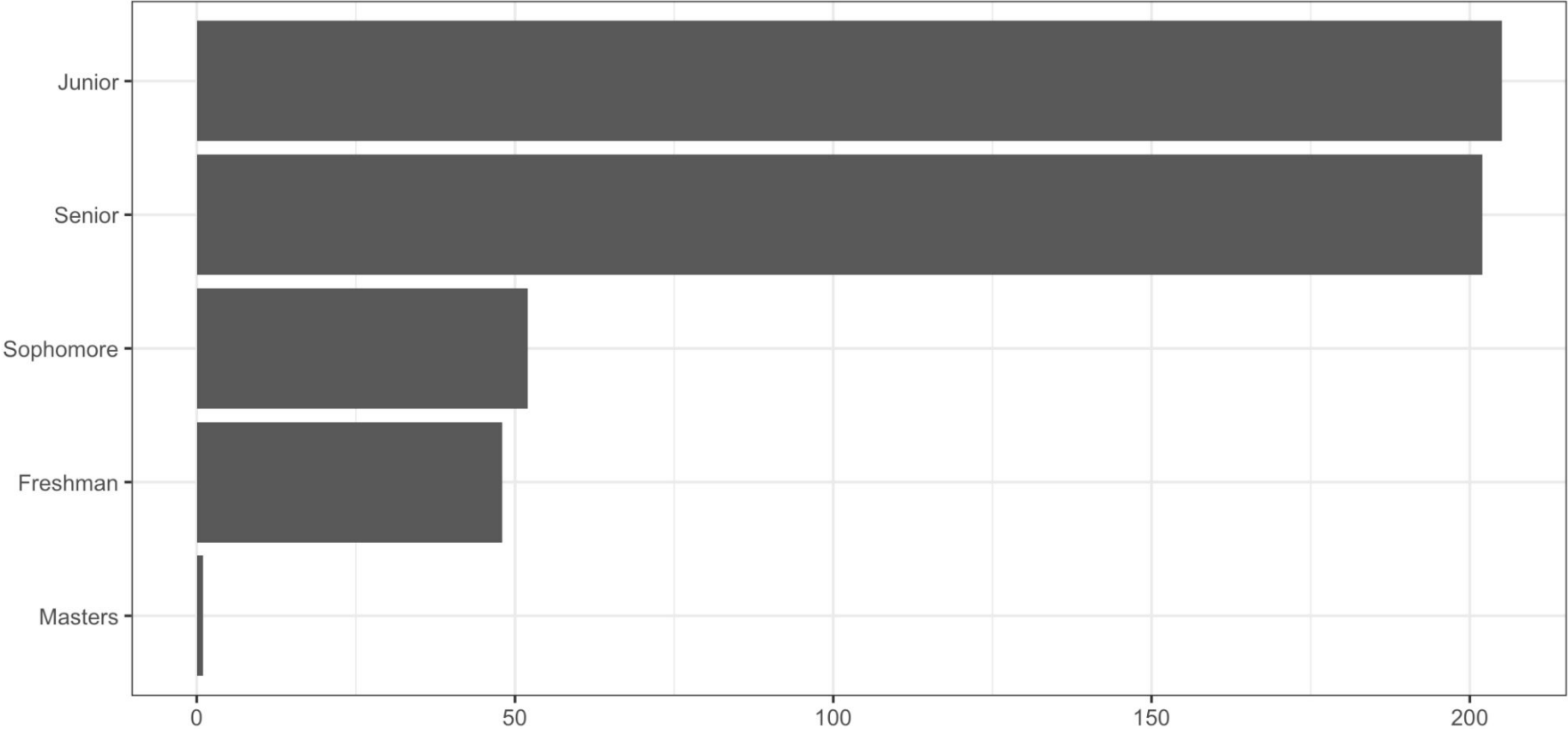
something else really awesome

**I'm excited to have
you all in COGS 108!**

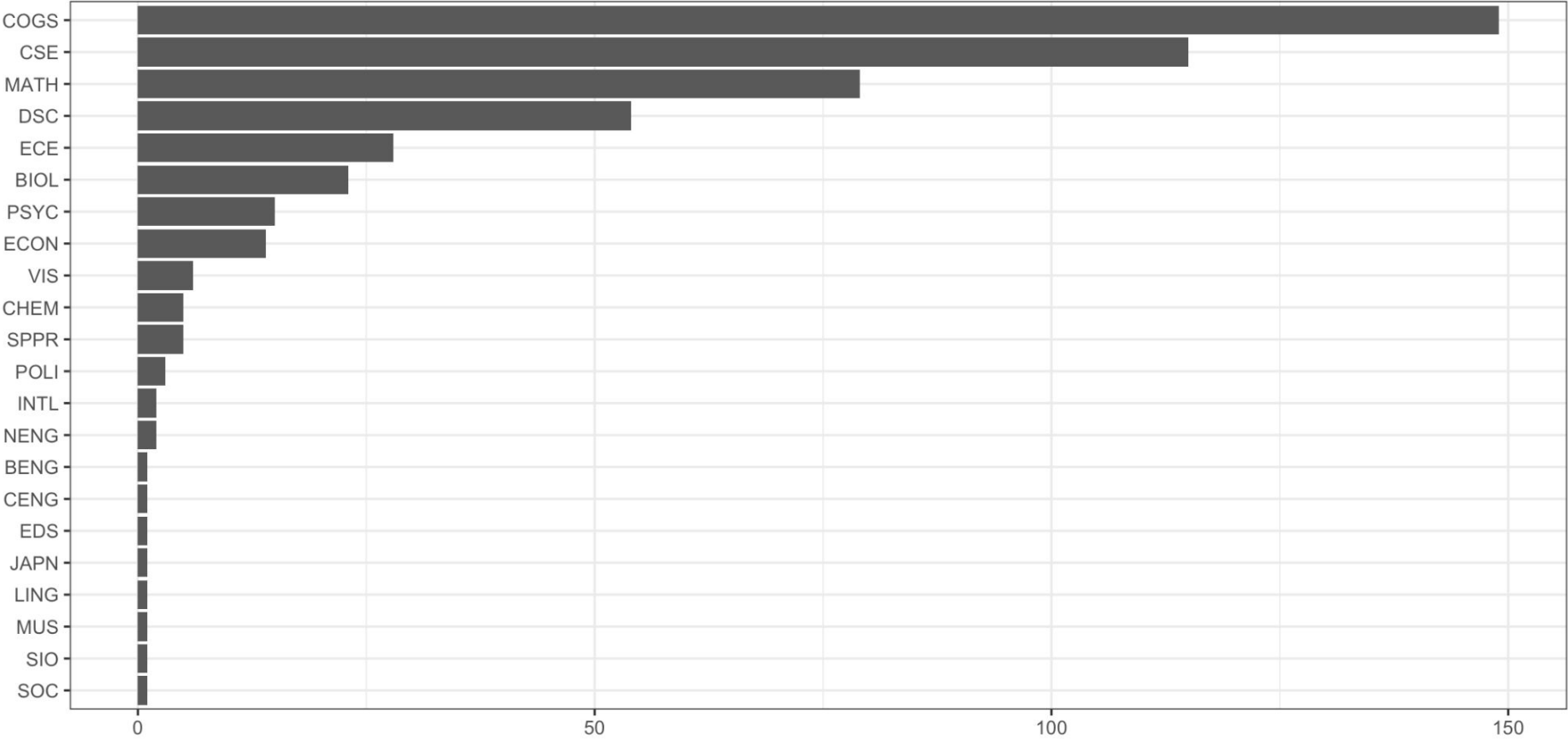
COGS 108: College



COGS 108: Year



COGS 108: Major



Survey (link also on canvas; #finaid)



due 11:59 PM
**Sunday after
week 2**

COGS 108 Student Survey (Fall 2026)



This survey must be completed by the Sunday following Week 2. It is being used as our #finaid survey.

This survey has three parts. The first part helps me get to know you better. It also collects data that will be anonymized and then used to demonstrate data science in class.

The second part collects your GitHub username. If you don't yet have one, you'll be directed where to go to get one.

The third part of the survey is collecting demographic information. THIS SECTION IS OPTIONAL - refusing to fill it out does NOT affect your grade. We use this to better understand how we're supporting/teaching students in this class.

If any of these data are used/displayed in class, the data will be anonymized. Please answer as truthfully as possible. How you respond will NOT affect how you do in this class. Also, many questions are NOT required. Please do not answer anything that makes you uncomfortable.

This form is automatically collecting emails from all respondents. [Change settings](#)